MEMORY FORENSICS REPORT

MFF v2 · Post-Volatility Analysis Framework

OVERALL SEVERITY: CRITICAL

Case ID	`case01_baseline_vs_case05_multi_attack`
Generated	`2026-04-18 19:37:05 UTC`
Baseline capture	`case01_baseline`
Attack capture	`case05_multi_attack`
Baseline path	`/MFF/cases/case01_baseline`
Attack path	`/MFF/cases/case05_multi_attack`

1. EXECUTIVE SUMMARY

Metric	Value	Metric	Value
New Processes (Attack Only)	16	Critical Risk Processes	8
Gone Processes (Baseline Only)	16	ATT&CK Techniques Observed	8
New Network Connections	139	ATT&CK Tactics Observed	3
Flagged Connections	4	IOCs Extracted	59

ATT&CK; Tactics Observed:

Defense Evasion · Discovery · Execution

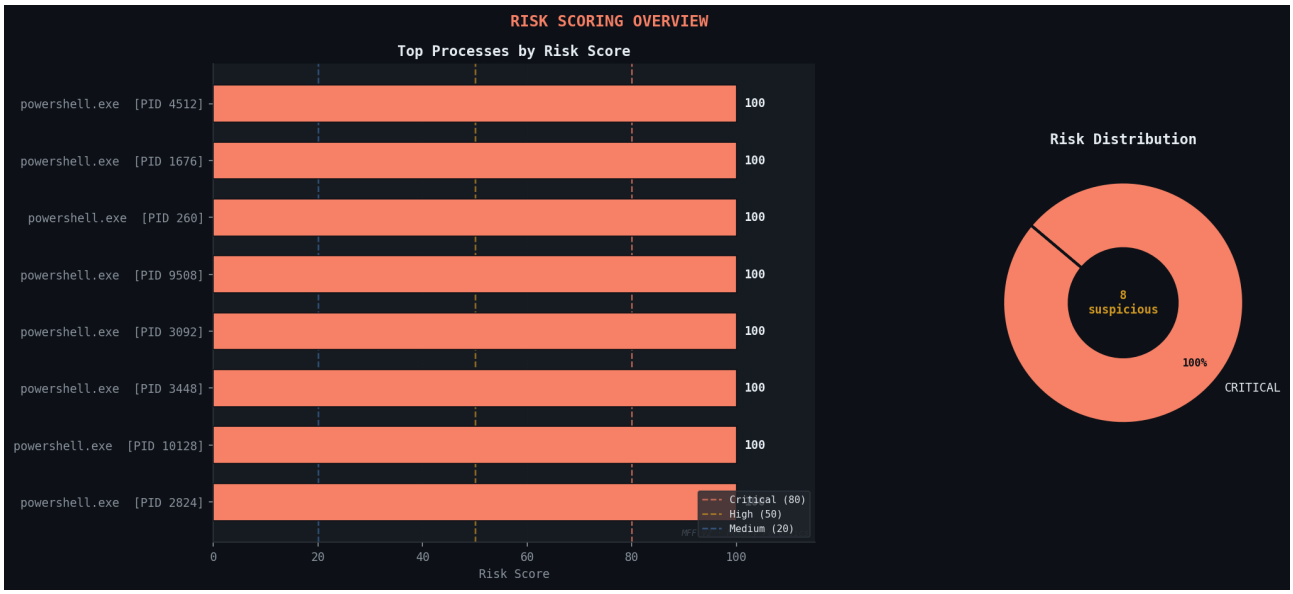
IOC Summary:

Type	Indicators (top 8)
IPv4	0.0.0.0 · 150.171.22.17 · 104.208.203.88 · 150.171.27.11 · 2.18.67.92 · 23.11.33.159

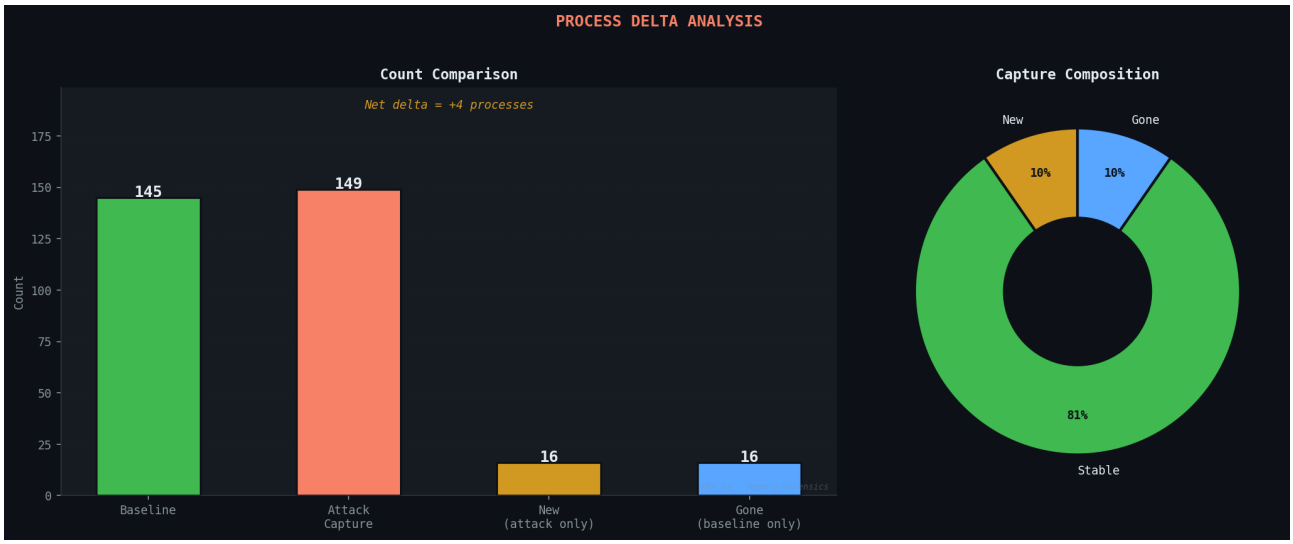
2. VISUALIZATIONS

2.1 Full Forensics Dashboard

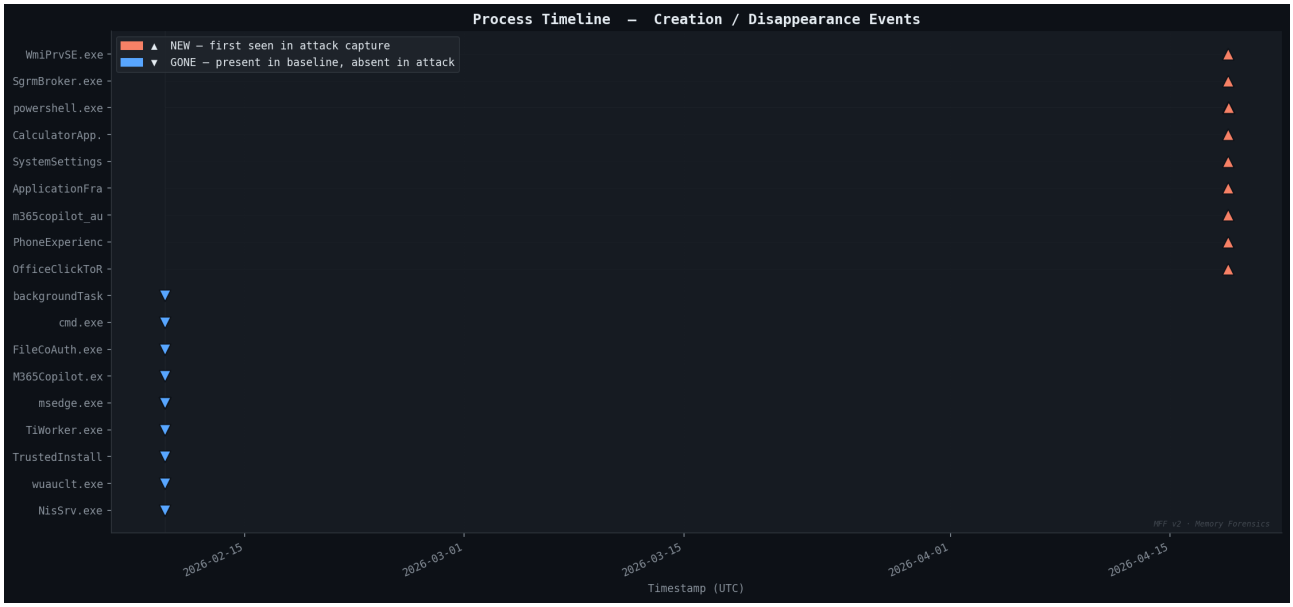
2.2 Risk Scoring Overview



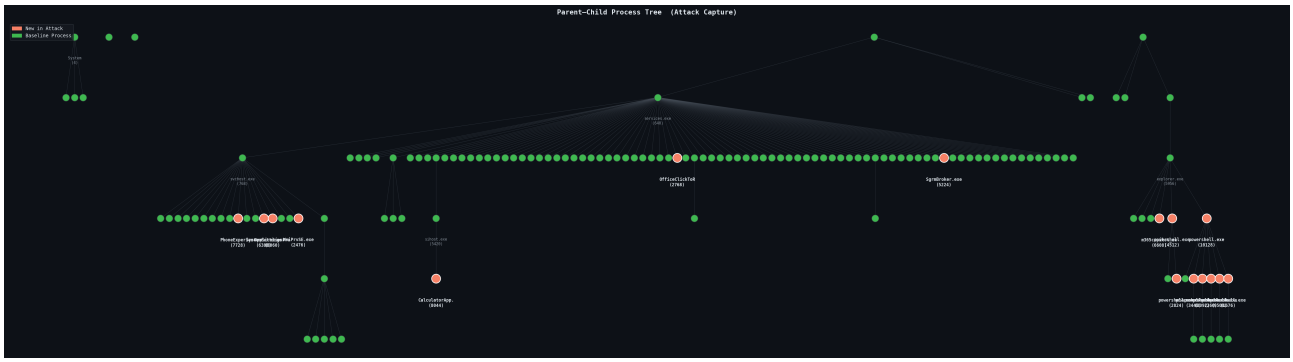
2.3 Process Delta Analysis



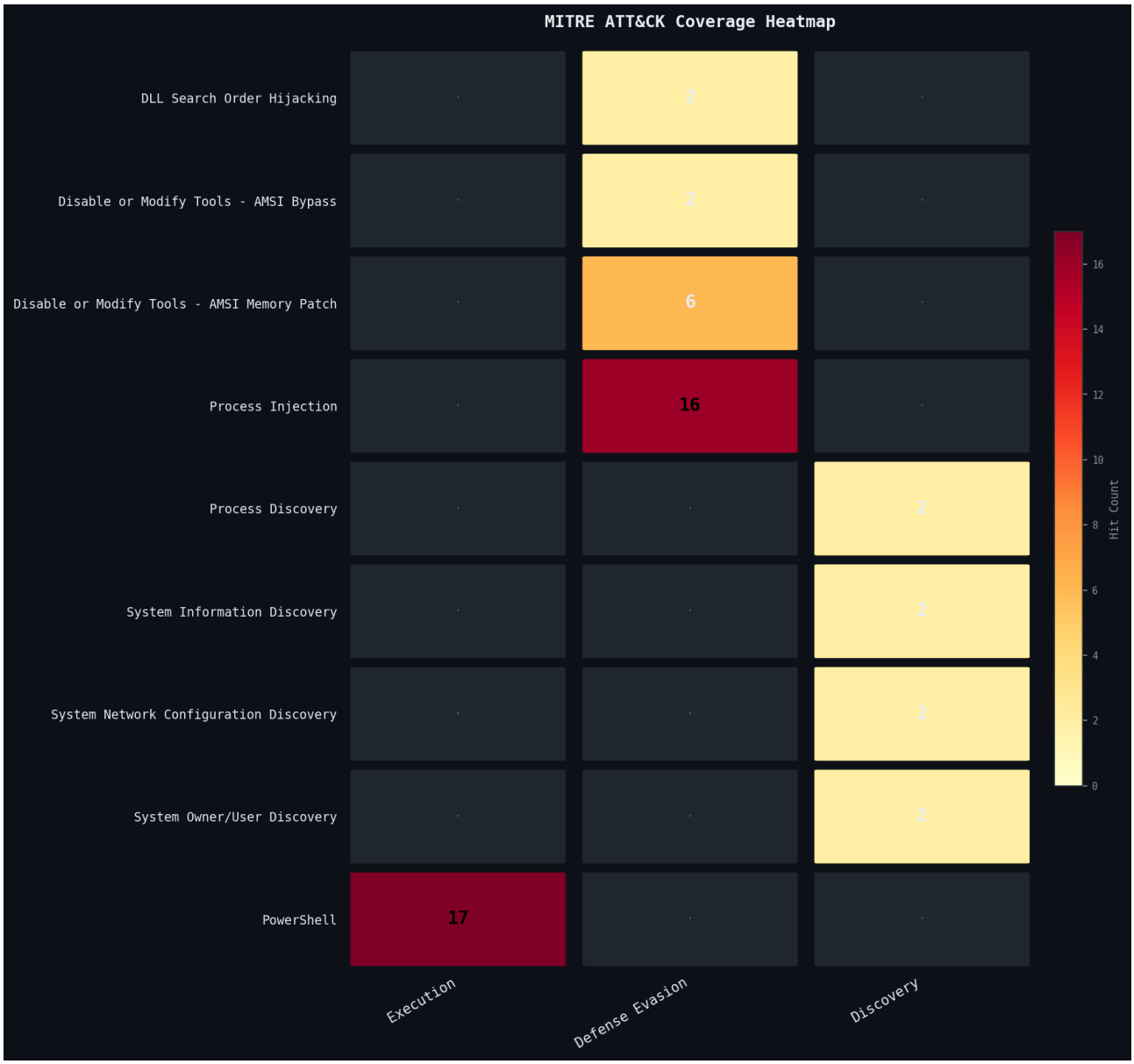
2.4 Process Timeline



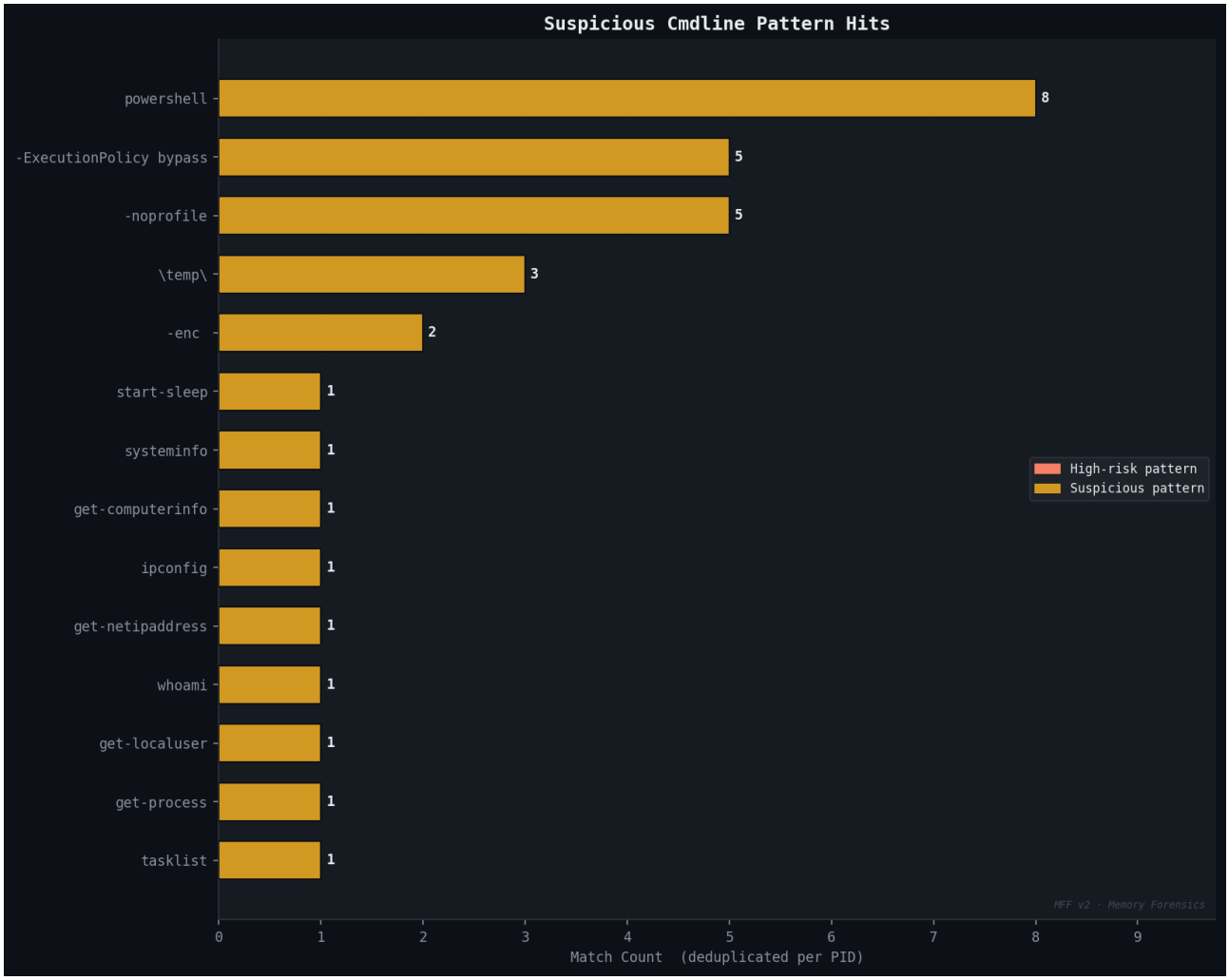
2.5 Parent-Child Process Tree



2.6 MITRE ATT&CK; Heatmap



2.7 Suspicious Cmdline Patterns



2.8 Malfind Memory Protections

3. RISK SCORE TABLE

Process	PID	Score	Level	Indicators		
powershell.exe	4512	100	CRITICAL	T1562.001	3 src	High
powershell.exe	1676	100	CRITICAL	T1562.001	5 src	Very High
powershell.exe	260	100	CRITICAL	T1562.001	5 src	Very High
powershell.exe	9508	100	CRITICAL	T1562.001	4 src	Very High
powershell.exe	3092	100	CRITICAL	T1562.001	5 src	Very High
powershell.exe	3448	100	CRITICAL	T1562.001	4 src	Very High
powershell.exe	10128	100	CRITICAL	T1562.001	3 src	High
powershell.exe	2824	100	CRITICAL	T1574.001	5 src	Very High

4. NEW PROCESSES (Attack Only)

ImageFileName	PID	PPID	CreateTime
OfficeClickToR	2768	640	2026-04-18 17:16:24.000000 UTC
PhoneExperienc	7728	768	2026-04-18 17:17:12.000000 UTC
m365copilot_au	6608	5956	2026-04-18 17:17:16.000000 UTC
SystemSettings	6388	768	2026-04-18 17:17:30.000000 UTC
ApplicationFra	8060	768	2026-04-18 17:17:30.000000 UTC
CalculatorApp.	8044	5420	2026-04-18 17:17:42.000000 UTC
powershell.exe	4512	5956	2026-04-18 17:17:58.000000 UTC
SgrmBroker.exe	5224	640	2026-04-18 17:18:15.000000 UTC
WmiPrvSE.exe	2476	768	2026-04-18 17:18:51.000000 UTC
powershell.exe	2824	4512	2026-04-18 17:19:32.000000 UTC
powershell.exe	10128	5956	2026-04-18 17:35:17.000000 UTC
powershell.exe	3448	10128	2026-04-18 17:36:02.000000 UTC
powershell.exe	3092	10128	2026-04-18 17:37:34.000000 UTC
powershell.exe	260	10128	2026-04-18 17:40:06.000000 UTC
powershell.exe	9508	10128	2026-04-18 17:40:14.000000 UTC
powershell.exe	1676	10128	2026-04-18 17:40:50.000000 UTC

5. MITRE ATT&CK; TECHNIQUE MATCHES

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	4512	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	2824	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	10128	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	3448	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	3092	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	260	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	9508	powershell.exe
T1055	Process Injection	Defense Evasion	PAGE_EXECUTE_READWRITE	1676	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	4512	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	2824	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	10128	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	3448	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	3092	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	260	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	9508	powershell.exe
T1055	Process Injection	Defense Evasion	EXECUTE_READWRITE	1676	powershell.exe
T1016	System Network Configuration Discovery	Discovery	ipconfig	260	powershell.exe
T1016	System Network Configuration Discovery	Discovery	get-netipaddress	260	powershell.exe
T1033	System Owner/User Discovery	Discovery	whoami	260	powershell.exe
T1033	System Owner/User Discovery	Discovery	get-localuser	260	powershell.exe
T1057	Process Discovery	Discovery	tasklist	260	powershell.exe
T1057	Process Discovery	Discovery	get-process	260	powershell.exe

Technique	TechniqueName	Tactic	MatchedKeyword	PID	Process
T1082	System Information Discovery	Discovery	systeminfo	260	powershell.exe
T1082	System Information Discovery	Discovery	get-computerinfo	260	powershell.exe
T1059.001	PowerShell	Execution	-enc	3448	powershell.exe
T1059.001	PowerShell	Execution	-enc	9508	powershell.exe
T1059.001	PowerShell	Execution	-noprofile	3448	powershell.exe
T1059.001	PowerShell	Execution	-noprofile	3092	powershell.exe
T1059.001	PowerShell	Execution	-noprofile	260	powershell.exe
T1059.001	PowerShell	Execution	-noprofile	9508	powershell.exe

6. SUSPICIOUS CMDLINE FINDINGS

ImageFileName	PID	MatchedPattern	Args
powershell.exe	4512	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
powershell.exe	2824	powershell	"C:\\Temp\\multiattack\\powershell.exe"
powershell.exe	10128	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
powershell.exe	3448	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	3092	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	9508	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	1676	powershell	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	3448	-ExecutionPolicy bypass	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	3092	-ExecutionPolicy bypass	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	-ExecutionPolicy bypass	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	9508	-ExecutionPolicy bypass	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	1676	-ExecutionPolicy bypass	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	3448	-enc	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	9508	-enc	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	3448	-noprofile	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	3092	-noprofile	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden

ImageFileName	PID	MatchedPattern	Args
powershell.exe	260	-noprofile	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	9508	-noprofile	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	1676	-noprofile	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	start-sleep	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	systeminfo	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	get-computerinfo	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	ipconfig	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	get-netipaddress	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	whoami	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	get-localuser	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	get-process	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	260	tasklist	"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe" -ExecutionPolicy Bypass -NoProfile -WindowStyle Hidden
powershell.exe	2824	\temp\	"C:\\Temp\\multiattack\\powershell.exe"

7. MALFIND — MEMORY INJECTION ARTEFACTS

Process	PID	Protection
powershell.exe	4512	PAGE_EXECUTE_READWRITE
powershell.exe	4512	PAGE_EXECUTE_READWRITE
powershell.exe	4512	PAGE_EXECUTE_READWRITE
powershell.exe	4512	PAGE_EXECUTE_READWRITE
powershell.exe	4512	PAGE_EXECUTE_READWRITE

Process	PID	Protection
powershell.exe	2824	PAGE_EXECUTE_READWRITE
powershell.exe	2824	PAGE_EXECUTE_READWRITE
powershell.exe	2824	PAGE_EXECUTE_READWRITE
powershell.exe	2824	PAGE_EXECUTE_READWRITE
powershell.exe	2824	PAGE_EXECUTE_READWRITE
powershell.exe	10128	PAGE_EXECUTE_READWRITE
powershell.exe	10128	PAGE_EXECUTE_READWRITE
powershell.exe	10128	PAGE_EXECUTE_READWRITE
powershell.exe	10128	PAGE_EXECUTE_READWRITE
powershell.exe	10128	PAGE_EXECUTE_READWRITE
powershell.exe	3448	PAGE_EXECUTE_READWRITE
powershell.exe	3448	PAGE_EXECUTE_READWRITE
powershell.exe	3448	PAGE_EXECUTE_READWRITE
powershell.exe	3448	PAGE_EXECUTE_READWRITE
powershell.exe	3448	PAGE_EXECUTE_READWRITE
powershell.exe	3092	PAGE_EXECUTE_READWRITE
powershell.exe	3092	PAGE_EXECUTE_READWRITE
powershell.exe	3092	PAGE_EXECUTE_READWRITE
powershell.exe	3092	PAGE_EXECUTE_READWRITE
powershell.exe	3092	PAGE_EXECUTE_READWRITE
powershell.exe	3092	PAGE_EXECUTE_READWRITE
powershell.exe	260	PAGE_EXECUTE_READWRITE
powershell.exe	260	PAGE_EXECUTE_READWRITE
powershell.exe	260	PAGE_EXECUTE_READWRITE
powershell.exe	260	PAGE_EXECUTE_READWRITE

7b. DLL ANALYSIS — T1574 / T1562 FINDINGS

12 DLL finding(s) — T1574 DLL Search Order Hijacking + T1562 AMSI Bypass evidence.

PID	Process	DLL	HijackType	Technique	RiskScore	Tactic
3092	powershell.exe	amsi.dll	AMSI_FILE_OUTPUT_DISABLED	T1562.001	90	Defense Evasion
1676	powershell.exe	amsi.dll	AMSI_FILE_OUTPUT_DISABLED	T1562.001	90	Defense Evasion
2824	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
4512	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
3092	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
260	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
10128	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
3448	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
9508	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
1676	powershell.exe	amsi.dll	AMSI_BYPASS_MEMORY_PATCH	T1562.001	80	Defense Evasion
1676	powershell.exe	amsi.dll	PROTECTED_DLL_USER_DIR	T1574.001	70	Defense Evasion
3092	powershell.exe	amsi.dll	PROTECTED_DLL_USER_DIR	T1574.001	70	Defense Evasion

[T1562.001] AMSI_FILE_OUTPUT_DISABLED — powershell.exe (PID=3092)

HIGHEST CONFIDENCE T1562.001+T1574.001: 'powershell.exe' (PID=3092) has amsi.dll from 'C:\\Temp\\mff_multi\\amsi.dll' (NOT System32). File output='disabled'. Replaced DLL confirmed.

[T1562.001] AMSI_FILE_OUTPUT_DISABLED — powershell.exe (PID=1676)

HIGHEST CONFIDENCE T1562.001+T1574.001: 'powershell.exe' (PID=1676) has amsi.dll from 'C:\\Temp\\mff_multi\\amsi.dll' (NOT System32). File output='disabled'. Replaced DLL confirmed.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=2824)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=2824) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=4512)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=4512) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=3092)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=3092) loads amsi.dll AND has 6 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=260)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=260) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=10128)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=10128) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

[T1562.001] AMSI_BYPASS_MEMORY_PATCH — powershell.exe (PID=3448)

HIGH CONFIDENCE T1562.001: 'powershell.exe' (PID=3448) loads amsi.dll AND has 5 PAGE_EXECUTE_READWRITE region(s). Standard pattern for PowerShell AMSI bypass via memory patching.

8. NETWORK ARTEFACTS

8a. Flagged Connections (suspicious ports)

TreeDepth	Offset	Proto	LocalAddr	LocalPort	ForeignAddr	ForeignPort	State	PID	Owner	Created	ImageFileName	SuspiciousPort	PortMeaning
0	0xa78863074050	TCPv4	10.0.2.15	139	0.0.0.0	0	LISTENING	4	System	2026-04-18 17:22:06.000000 UTC	System	139	NetBIOS
0	0xa78641ba050	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	888	svchost.exe	2026-04-18 17:16:21.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xa78641bacb0	TCPv4	0.0.0.0	135	0.0.0.0	0	LISTENING	888	svchost.exe	2026-04-18 17:16:21.000000 UTC	svchost.exe	135	DCOM/RPC
0	0xa78641bacb0	TCPv6	::	135	::	0	LISTENING	888	svchost.exe	2026-04-18 17:16:21.000000 UTC	svchost.exe	135	DCOM/RPC

9. IOC DETAIL TABLE

Type	Value	Source	Count
FilePath	C:\\Windows\\system32\\svchost.exe	cmdline	46
FilePath	C:\\Windows\\System32\\svchost.exe	cmdline	29
FilePath	C:\\Program	cmdline	13
FilePath	??\\C:\\Windows\\system32\\conhost.exe	cmdline	7
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.Windows.Search_cw5n1h2txyewy\\LocalState\\EBWebView	cmdline	7
FilePath	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe	cmdline	7
FilePath	Microsoft\\EdgeWebView\\Application\\146.0.3856.109\\msedgewebview2.exe	cmdline	7
FilePath	C:\\Windows\\System32\\RuntimeBroker.exe	cmdline	5
FilePath	C:\\ProgramData\\Microsoft\\Windows	cmdline	2
FilePath	C:\\Temp\\mff_multi\\case05_rwx_amsi.ps1	cmdline	2
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.Search_cw5n1h2txyewy\\SearchApp.exe	cmdline	2
FilePath	C:\\Windows\\system32\\DllHost.exe	cmdline	2
FilePath	Microsoft	cmdline	2
FilePath	Windows	cmdline	2
FilePath	system32\\csrss.exe	cmdline	2
FilePath	C:\\Temp\\multiattack\\powershell.exe	cmdline	1
FilePath	C:\\Users\\lucif\\AppData\\Local\\Packages\\Microsoft.Windows.Search_cw5n1h2txyewy\\LocalState\\EBWebView\\Crashpad	cmdline	1
FilePath	C:\\Windows\\Explorer.EXE	cmdline	1
FilePath	C:\\Windows\\ImmersiveControlPanel\\SystemSettings.exe	cmdline	1
FilePath	C:\\Windows\\System32\\SecurityHealthSystray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxService.exe	cmdline	1
FilePath	C:\\Windows\\System32\\VBoxTray.exe	cmdline	1
FilePath	C:\\Windows\\System32\\mousocoreworker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\oobe\\UserOOBEBroker.exe	cmdline	1
FilePath	C:\\Windows\\System32\\spoolsv.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\\StartMenuExperienceHost.exe	cmdline	1
FilePath	C:\\Windows\\SystemApps\\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\\TextInputHost.exe	cmdline	1

Type	Value	Source	Count
FilePath	C:\\Windows\\SystemApps\\ShellExperienceHost_cw5n1h2txyewy\\ShellExperienceHost.exe	cmdline	1
FilePath	C:\\Windows\\system32\\ApplicationFrameHost.exe	cmdline	1
FilePath	C:\\Windows\\system32\\SearchIndexer.exe	cmdline	1

10. METHODOLOGY

Acquisition: VBoxManage debugvm dumpvmcore — cold hypervisor-level acquisition. No kernel module or guest agent. Forensically sound. SHA256 + MD5 hash recorded on host immediately post-acquisition.

Volatility 3: Memory parsed using Volatility 3: pslist, pstree, cmdline, dlllist, malfind, netscan, threads. Exported as CSV (engine) and JSONL (audit trail).

Process diff: Baseline vs attack comparison by process name (ImageFileName). PID-based diff rejected — PIDs are reassigned on every reboot and produce false-positive churn exceeding 90% of all processes.

Risk scoring: Transparent rule-based scoring (no ML). RWX memory region +60 · Suspicious cmdline pattern +40 · Known-malicious process name +50 · Suspicious DLL load (T1574) +30 · Anomaly name bonus +15. Thresholds: CRITICAL ≥80 HIGH ≥50 MEDIUM ≥20 LOW <20.

Malfind filter: Known-clean JIT processes excluded (browsers, Defender, .NET, OneDrive). Self-match false positives suppressed (e.g. lsass.exe matching itself). Only genuinely suspicious PAGE_EXECUTE_READWRITE regions retained.

DLL analysis: windows.dlllist CSV analysed for T1574 DLL Search Order Hijacking. Protected DLLs (amsi.dll, version.dll, cryptbase.dll etc.) checked for loads from non-System32 paths. Findings merged into ATT&CK; tagged output.

ATT&CK; mapping: 30+ rule-based signatures across 8 MITRE ATT&CK; tactics. Keyword matching on cmdline args, process names, DLL paths, and network artefacts. T1574-specific patterns include: copy-item, amsi.dll, invoke-atomictest.

IOC extraction: Regex-based extraction: IPv4 (excluding RFC1918 private ranges), domains, URLs, MD5/SHA1/SHA256 hashes, Windows file paths.

Framework: MFF v2 — Python Post-Volatility Memory Forensics Framework. FYP project — University of Roehampton 2026. Tool chain: VBoxManage → Volatility 3 → MFF v2 → HTML + PDF reports. Modules: comparison_engine_v2, dll_analysis, mitre_tagger, network_ioc, process_tree, export_alert, report_generator.

Risk scoring: Transparent additive model — no ML. Each evidence category contributes once. Weights calibrated to specificity (ACPO P2). CRITICAL>=80, HIGH>=55, MEDIUM>=25.

Standards: ACPO Good Practice Guide for Digital Evidence (Principles 1-4). NIST SP 800-86 Integrating Forensic Techniques into Incident Response. ISO/IEC 27037:2012 Guidelines for Identification, Collection and Preservation. College of Policing Digital Forensics Guidance. CPS Disclosure Manual.